

Zeyad El-Sayed

AI / ML Engineer — Machine Learning Engineer

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Military Service: Completed

TECHNICAL SUMMARY

AI/ML Engineer and Machine Learning Engineer focused on LLM applications, **RAG**, and NLP systems. Built TesseractRAG, a multilingual RAG project featuring hybrid retrieval, LLM routing, and modular backend architecture for knowledge-base question answering. Experienced in fine-tuning large language models using PEFT techniques (LoRA, QLoRA), developing transformer-based deep learning systems, and deploying AI services with FastAPI and Docker. Open to AI Engineer, ML Engineer, Applied AI, and LLM Engineering roles.

SKILLS

Programming Languages: Python, C++, SQL

LLM & RAG Engineering: LLM Applications, Hybrid Retrieval, Prompt Engineering, Fine-Tuning (PEFT), pgvector, BM25, Cross-Encoder Reranking

ML Frameworks: PyTorch, Keras, HuggingFace Transformers, Scikit-learn

Deep Learning: Transformers, CNNs, RNNs, LSTMs, Autoencoders, GANs

Data & Scientific Computing: Pandas, NumPy, OpenCV

Backend & Infrastructure: FastAPI, REST APIs, PostgreSQL, Redis, Docker

MLOps & Deployment: GitHub Actions (CI/CD), Model Deployment, Railway, Cloudflare Workers

Data Engineering & BI: Tableau, Power BI, Excel

Developer Tools: Git, VS Code, Linux

EXPERIENCE

Information Technology Institute (ITI) — Business Intelligence Trainee Jul 2023 – Sep 2023 *Egypt*
• Designed interactive dashboards using **Tableau** and **Power BI**, translating raw datasets into actionable business insights.
• Gained foundational experience in ETL pipelines, dimensional modeling, and SQL-based data warehouse design.

PROJECTS

TesseractRAG | *FastAPI, PostgreSQL, pgvector, BM25, Redis, Docker*

📄 GitHub | 🌐 Live Demo

- Rebuilt v2 from scratch after v1's flat architecture proved unextendable – redesigned around abstract base classes and dispatcher routing so any component (embedder, retriever, generator, database,etc..) is swappable by adding one class, touching nothing else.
- Implemented hybrid BM25 + pgvector retrieval fused via RRF, improving multilingual retrieval relevance with 1–3s response latency through automatic semantic-only fallback for Arabic/English cross-language queries.
- Designed a 3-zone LLM routing layer (SmartGuard) tracking daily Groq quota in Redis – Groq-only below 80%, round-robin for graceful degradation to 100%, HuggingFace fallback beyond – with query rewriting optimized separately for semantic and BM25 .
- Integrated structured logging, health-check endpoints, and provider fallback logic to improve observability, debugging, and service reliability during quota exhaustion or provider failure.
- Replaced v1's JSON + serialized FAISS with PostgreSQL + pgvector – eliminating partial-write corruption risk and gaining ACID semantics and versioned migrations via Alembic.
- Switched reranker from **bge-reranker-base** to **BAAI/bge-reranker-v2-m3** – a multilingual cross-encoder – after the monolingual model failed on Arabic/English mixed retrieval.
- Added role-based analytics layer – admin dashboard (users, sessions, docs, avg score, per-session eval drill-down) and a per-user stats panel in the chat UI – via JWT-scoped FastAPI endpoints on the existing SQLAlchemy models.

Catch the AI | *Python, Django, React, PostgreSQL, ViT, DeBERTa, Wav2Vec 2.0*

📄 GitHub | 🌐 Website

- Built ViT and EfficientNet models to detect AI-generated images with **96%** accuracy for media verification.
- Built an NLP-based detector (DeBERTa, RoBERTa) with **95%** accuracy in spotting AI-generated text.
- Implemented Wav2Vec 2.0 for speech detection, achieving **90%** accuracy in identifying AI-generated audio.

Fine-Tuning LLMs | *PyTorch, PEFT, HuggingFace Transformers, NLP*

📄 GitHub | 🌐 HF Spaces

- Curated a fine-tuning repository and deployed models on Hugging Face Spaces for text classification, generation, and summarization.
- Applied LoRA and QLoRA PEFT techniques to reduce trainable parameters by up to 99% while maintaining performance.
- Evaluated models using ROUGE-L, BLEU, and additional metrics, achieving 95%+ performance across tasks.

LLMs from Scratch | *PyTorch, NLP, Transformers*

📄 GitHub

- Implemented a Large Language Model from scratch – tokenization, multi-head self-attention, positional encodings, and full transformer blocks in pure PyTorch.
- Built end-to-end training and evaluation pipelines, deepening understanding of modern NLP architecture internals.

EDUCATION

Benha University B.S. in Computers & Artificial Intelligence

GPA: 3.4 / 4.0 — Oct 2020 - Jun 2024

LANGUAGES

Arabic: Native | **English:** Professional working proficiency (reading, writing, technical communication)